

1 Appetizer: Op-Amps and Feedback

1.1 Designing a reliable op-amp from unreliable parts

An amplifier A satisfies the relationship $V_o = AV_i$. I could draw this as

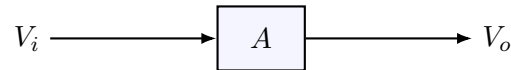


Figure 1.1: An amplifier A takes an input voltage V_i and produces an output voltage $V_o = AV_i$. What can we do if A is large but very uncertain?

Suppose I am in the op-amp business, and my manufacturer can make these very cheaply, but with extreme variation in the gain A . More specifically, suppose that any given amplifier A can only be guaranteed to satisfy $A \in [1000, 100000]$. However, they can also produce attenuators $k \in [0, 1]$, such that $V_o = kV_i$, very accurately and cheaply. How can I use these two components, an inaccurate high-gain amplifier and an accurate attenuator, to produce an accurate amplifier?

The answer is to use **feedback**, in which we take the output V_o , and feed it back into the input of the system.

1.2 Open-loop vs. closed-loop

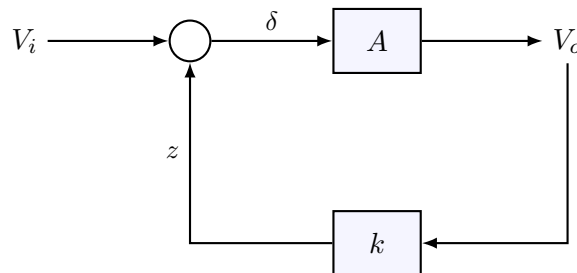


Figure 1.2: An example of closed-loop system induced by negative feedback. The output V_o of the uncertain amplifier A is passed through the accurate attenuator k , and then subtracted from the input V_i . The resulting difference δ is then input to the amplifier. This feedback of the output back into the input of the system leads to an implicit equation in V_o .

Consider the circuit illustrated in Fig. 1.2, in which the attenuator k is connected in **negative feedback** with the amplifier A . This results in a **closed loop** system in which the output V_o depends implicitly on itself. In contrast, the system in Fig. 1.1 is called **open-loop**, as no such interconnection from output to input exists.

Let's unpack the behavior of the closed-loop system a bit more carefully by labeling the internal signals and writing out their relationships

$$V_o = A\delta \quad (1.1)$$

$$z = kV_o \quad (1.2)$$

$$\delta = V_i - z. \quad (1.3)$$

These three equations and three unknowns can be solved for in terms of A , k , and V_i . We are particularly interested in the input/output relationship $V_i \rightarrow V_o$. We first note that eliminating z and δ from the above yields the *implicit* expression for the output voltage $V_o = A(V_i - kV_o)$. If $Ak \neq -1$, we can solve for V_o as

$$V_o = \frac{A}{1 + Ak} V_i, \quad (1.4)$$

💡 *Key Result 1.1*

If $Ak \neq -1$, we say that the feedback interconnection is **well posed**. It is not well posed otherwise.

Exercise 1.1. What does it physically mean if $Ak = -1$? Algebraically, what happens to the system of equations?

Notice that for *any* A sufficiently large, we have that $V_o/V_i \approx 1/k$, and thus we achieve an effective amplifier with gain $1/k$ even if the value of A is not known accurately at all.

Example 1.2. Suppose we want an amplifier to with effective gain of 10, and so we set $k = 0.1$. Then:

$$\begin{aligned} A = 1000 &\Rightarrow \frac{1000}{101} \approx 9.9 \\ A = 100000 &\Rightarrow \frac{100000}{10001} \approx 9.999 \end{aligned}$$

Remark 1.3. We achieve 1% accuracy or better over three orders of magnitude!

Remark 1.4. This is a stylized version of Black's feedback amplifier [Bla34]. We omit the details, but a similar "robustness through feedback" phenomenon can be achieved by appropriately connecting resistors in feedback with a modern op amp. See [ÅM21, Ch. 3.3] for more details

1.3 What's the catch?

Waterbed effect A key fact from robust control is that when you introduce robustness into a system, it always leads to increased fragility elsewhere. This is known as a **waterbed effect**, **design constraint**, or **conservation law** (see [DFT92, Ch. 6] for more details). Here, we see a very simple illustration of this. In particular, while we are robust to extreme uncertainty in A , we become very sensitive to uncertainty in k .

Example 1.5. We now make the robustness claim above precise. For a function f of a

variable x , define the *logarithmic sensitivity* of f with respect to x as

$$\mathcal{S}_x^f := \frac{\partial \ln f}{\partial \ln x}, \quad (1.5)$$

the fractional change in f induced by a fractional change in x . Since $\partial \ln f / \partial x = (1/f) \partial f / \partial x$ by the chain rule, and $\partial x / \partial \ln x = x$, a second application of the chain rule gives the equivalent, directly computable expression

$$\mathcal{S}_x^f = \frac{\partial \ln f}{\partial x} \cdot \frac{\partial x}{\partial \ln x} = \frac{x}{f} \frac{\partial f}{\partial x}. \quad (1.6)$$

Apply this with $f = V_o/V_i = A/(1 + Ak)$. Holding k fixed and differentiating with respect to A ,

$$\mathcal{S}_A^{V_o/V_i} = \frac{1}{1 + Ak} =: S, \quad (1.7)$$

the classical *sensitivity function*. For $Ak \gg 1$ – the regime of interest here – S is small, so a given fractional change in A produces a much smaller fractional change in the closed-loop gain. Holding A fixed and differentiating with respect to k ,

$$\mathcal{S}_k^{V_o/V_i} = -\frac{Ak}{1 + Ak} =: -T, \quad (1.8)$$

where $T := Ak/(1 + Ak)$ is the classical *complementary sensitivity function*. For $Ak \gg 1$, $\mathcal{S}_k^{V_o/V_i} \rightarrow -1$: a given fractional change in k produces an essentially equal and opposite fractional change in the closed-loop gain.

The two sensitivities are governed by the algebraic identity $S + T = 1$ (see [DFT92, Ch. 6]), which gives

$$\mathcal{S}_A^{V_o/V_i} - \mathcal{S}_k^{V_o/V_i} = S + T = 1. \quad (1.9)$$

This is the precise statement of the conservation law claimed above: sensitivity cannot be reduced with respect to A without an equal amount appearing (in this case with reversed sign) with respect to k .

Beware of delays The above model is overly simple, assuming instantaneous response, and constant inputs (and hence outputs). Here we consider a simple refinement, in which we model the system as evolving in discrete time. We introduce an **actuation delay** d to model the lag in the feedback loop:

$$V_o(t) = AV_i(t) - AkV_o(t - d). \quad (1.10)$$

Unrolling the recursion, we see that

$$\begin{aligned} V_o(t) &= AV_i(t) - Ak(AV_i(t - d) - AkV_o(t - 2d)) \\ &= \dots \\ &= A \sum_{j=0}^{\infty} (-Ak)^j V_i(t - jd). \end{aligned} \quad (1.11)$$

Thus the current output voltage is a linear superposition of suitably attenuated ($|Ak| < 1$) or amplified ($|Ak| > 1$) past inputs.

If inputs $|V_i(t)| \leq V_{\max}$ for all $t = 0, 1, \dots$, then equation (1.11) converges if and only if $|Ak| < 1$.

Furthermore, if $V_i(t) \equiv \bar{V}_i$, i.e., if the input voltage is constant, then

$$V_o(\infty) = \frac{A}{1 + Ak} \bar{V}_i, \quad (1.12)$$

i.e., we converge to the same input/output relationship as before!

This is also called the DC-gain of the system.

Remark 1.6. If you are familiar with transfer functions, this also follows immediately from taking the z -transform of equation (1.11). The condition $|Ak| < 1$ ensures that all poles of the resulting system are within the unit disk.

Our discussion above allows us to conclude that

- To be robust to actuation delay, we need $|Ak| < 1$.
- To be robust to uncertainty in A (for a fixed k), need $A \gg 1$.

💡 *Key Result 1.2*

This can be viewed as another sort of conservation law, wherein one cannot be simultaneously robust to actuation delays and uncertainty in A .

1.4 Things to keep in mind

Some simple things to keep in mind moving forward:

- Not all parts of a system need to be learned accurately!
- Very different open loop systems can look very similar in closed loop under the *same* feedback law!
- Robustness somewhere introduces fragility elsewhere.
- Unmodeled dynamics (e.g., actuation delays) can cause real trouble.

1.5 Further reading

For a first, gentle pass over the same territory (feedback amplifiers, the robustness argument), see Åström and Murray [ÅM21] and Black [Bla34], the original feedback-amplifier paper this whole story descends from. Ben Recht's *arg min* posts Recht [Rec24; Rec26] served as inspiration for this appetizer. For the “catch” in §1.3 – why robustness to one thing is never free – Doyle, Francis, and Tannenbaum [DFT92], Chapter 6 (“Design Constraints”), works out the underlying algebraic and analytic limits (the $S + T = 1$ identity and its consequences) in full generality; Stein [Ste03] is the classic essay-length treatment of the same conservation-of-robustness theme, argued from the unstable-plant side.

References

- [ÅM21] Karl J. Åström and Richard M. Murray. *Feedback Systems: An Introduction for Scientists and Engineers*. 2nd. Princeton University Press, 2021.
- [Bla34] Harold S. Black. “Stabilized Feedback Amplifiers”. In: *Bell System Technical Journal* 13.1 (1934), pp. 1–18.
- [DFT92] John C. Doyle, Bruce A. Francis, and Allen R. Tannenbaum. *Feedback Control Theory*. New York: Macmillan Publishing Co., 1992.
- [Rec24] Benjamin Recht. *The Soothing Warmth of Vacuum Tubes*. Feb. 2024. URL: <https://www.argmin.net/p/the-soothing-warmth-of-vacuum-tubes> (visited on 08/18/2026).
- [Rec26] Benjamin Recht. *Sweet Spots for Analysis*. Jan. 2026. URL: <https://www.argmin.net/p/sweet-spots-for-analysis> (visited on 08/18/2026).
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